Safwat Ali Khan

@ Email in LinkedIn Github Website

Education

GEORGE MASON UNIVERSITY
PhD in Computer Science
Expected 2026
CGPA 3.88/4.00

UNIVERSITY OF DHAKA
BSc in Computer Science
2010-2014 Bangladesh

Graduate Courseworks

Software Testing
Algorithm
UI Design and Development
Deep Learning
Natural Language Processing

Skills

MACHINE LEARNING
Prompt Engineering • Fine-tuning LLMs • OpenCV • tensorflow • pyTorch • CLiP • Transformer

AUTOMATION TOOL:
UIAutomator • Selenium webdriver • JMeter • Google Monkey • APE

LANGUAGES
Python • Java • C/C++ • JavaScript

Research Talk

FRAUNHOFER USA
Jan, 2023 Maryland, USA
Leveraging Machine Learning to bypass Android UI tarpits

ICST DOCTORAL SYMPOSIUM
Apr, 2025 Napoli, Italy
Advancing Mobile UI Testing by Learning Screen Usage Symantics

Mentorship

ASSIP 2020, 2022
Co-mentor for Aspired Scientist Summer Internship Program

Service

CO-REVIEWER
FSE 2024, ICSE 2023, SANER 2023, ASE 2022

STUDENT VOLUNTEER
ICST 2024

Research Publications

1. "AURORA: Navigating UI Tarpits via Automated Neural Screen Understanding" — **Safwat Ali Khan**, Wenyu Wang, Yiran Ren, Bin Zhu, Jiangfan Shi, Wing Lam, Kevin Moran. In ICST 2024.
2. "Testing Practices, Challenges, and Developer Perspectives in Open-Source IoT Platforms" — Daniel Rodriguez-Cardenas, **Safwat Ali Khan**, Prianka Mandal, Adwait Nadkarni, Kevin Moran, Denys Poshyvanyk. In ICST 2025.
3. "On Using GUI Interaction Data to Improve Text Retrieval-based Bug Localization" — Junayed Mahmud, Nadeeshan De Silva, **Safwat Ali Khan**, Seyed Hooman Mostafavi, S M Hasan Mansur, Oscar Chaparro, Andrian Marcus, Kevin Moran. In ICSE 2023.
4. "AVGUST: Automating Usage-based Test Generation from Videos of App Executions" — Yixue Zhao, Saghar Talebipour, Kesina Baral, Hyojae Park, Leon Yee, **Safwat Ali Khan**, Yuriy Brun, Nenad Medvidovic, Kevin Moran. In ESEC/FSE 2022.

Current Projects

- Developing and evaluating techniques to analyze automated input generation traces to **detect use cases** in mobile UI testing.
- Identifying usability issues in mobile apps and **redesigning UIs** based on user needs to improve overall user experience.

Fellowship

- CS Research Initiation Award — 2020
- Sustainability Summer Research Fellowship — 2024

Work Experience

GRADUATE ASSISTANT
George Mason University
Sep 2019 – Present Fairfax, VA, USA
Experience in academic research and teaching focusing software testing, data analysis, and program development.

- Conducting research on **automated mobile UI testing**, developing learning-based techniques to detect use cases from large-scale execution traces and analyzing limitations in smart home app testing.
- Designing and evaluating **IR and LLM-based methods** to identify user-facing use cases within automated exploration traces.
- Serving as a **Teaching Assistant**, mentoring students in Low-Level Programming (C, Java), Database Systems, and Data Structures courses.

INDUSTRY EXPERIENCE
Therap Services Ltd.
Jan 2018 – Jul 2019 Dhaka, Bangladesh
Relisource Technologies Ltd.
May 2015 – Dec 2017 Dhaka, Bangladesh
mPower Social Enterprises Ltd.
Aug 2014 – Apr 2015 Dhaka, Bangladesh
Software Quality Assurance Engineer with **5+ years of experience** in manual and automated testing of web and mobile applications, contributing to large-scale systems across diverse domains.

- Supported projects delivering digital solutions for underserved populations in Bangladesh.
- Worked on web-based learning platforms for healthcare professionals and health tracking modules for developmentally disabled individuals.